

Six-dimensional extensions of the five-dimensional kissing configurations Q_5 and R_5 : a proof of a conjecture of Cohn and Rajagopal

with rigidity and maximality certificates for D_5 , L_5 , Q_5 , R_5

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Abstract

We prove Conjecture 3.1 of Cohn and Rajagopal: no six-dimensional kissing configuration with Szöllősi’s Q_5 or Cohn–Rajagopal’s R_5 as a cross section contains 72 or more points; in fact at most 70. The proof combines a reduction lemma (any such cross section is central and the configuration splits into two independent layers), a norm-coupling inequality showing each layer projects to a spherical code in S^4 with inner products at most $1/5$, the sharp linear-programming bound $A(5, 1/5) = 16$ with its equality analysis à la Delsarte–Goethals–Seidel [10], and the exact computation that the deep-hole graphs of Q_5 and R_5 have clique number 12 (versus 16 for D_5, L_5). We also construct explicit 64-point six-dimensional kissing configurations with Q_5 and R_5 cross sections and conjecture that 64 is optimal. As supporting results, obtained by exact computation, we show that the four known 40-point kissing configurations in $\mathbb{R}^5$ — the D_5 configuration, Leech’s L_5 , Q_5 , and R_5 — are infinitesimally jammed, hence jammed (for Q_5 this answers a question of Szöllősi; for D_5, L_5 it reproduces results of Cohn, Jiao, Kumar, and Torquato), and that no 41st point can be adjoined to any of the four; the second fact is an input to the main argument. All certificates are exact data (rational, or in $\mathbb{Q}(\sqrt{3})$ for the 64-point extensions), verified by independent computations, and accompany the paper.

1 Introduction

The kissing number $\kappa(5)$ satisfies $40 \leq \kappa(5) \leq 44$ [1, 5], and 40 is widely conjectured optimal. For decades, two non-isometric 40-point kissing configurations in $\mathbb{R}^5$ were known — the minimal vectors of the root lattice D_5 , and a non-lattice configuration of Leech [2] — and both were conjectured to be optimal; it was believed that no others existed [3, 4]. Szöllősi [6] found a third

*Computer assistance. All code, certificates and the first draft of this note were produced with Claude Fable 5 (Anthropic) under the author’s direction (with Claude Opus 5 and Claude Sonnet 5 as subagents for the verifier implementations and literature retrieval, respectively). The results were then independently re-verified from the committed code in fresh sessions with Claude Fable 5.1, Gemini Pro and MiniMax M3, and by the exact-arithmetic checks described in Section 5. The author is responsible for this publication, the repository and this text; the mathematics was developed by the systems named above and verified as described in Section 5.

configuration Q_5 , refuting that belief, and Cohn and Rajagopal [7] subsequently found a fourth, R_5 .

We work at squared norm 2: following the usage of [7], a *kissing configuration* is a finite $X \subset \mathbb{R}^d$ with $\langle x, x \rangle = 2$ and $\langle x, y \rangle \leq 1$ for $x \neq y$ (no size restriction is implied; the term refers to the touching condition, not to maximality); *contacts* are pairs with $\langle x, y \rangle = 1$. All four configurations have purely rational coordinates in this normalization [7, Table 2.2]; each has exactly 240 contacts forming a 12-regular graph.

Cohn and Rajagopal [7, §3] extend D_5 and L_5 to six dimensions by placing 16-element sets of pairwise compatible deep holes in two layers above and below the cross section; this yields 72-point six-dimensional kissing configurations, among them those of E_6 and of Leech. For Q_5 and R_5 the same construction breaks down — no 16 deep holes are pairwise compatible (Lemma 4) — and they conjectured [7, Conjecture 3.1] that no six-dimensional kissing configuration with Q_5 or R_5 as a cross section contains 72 or more points. Our main result proves this with the stronger bound 70.

Theorem 1 (Conjecture 3.1 of [7]). *No six-dimensional kissing configuration containing a similarity copy of Q_5 or R_5 has more than 70 points. In particular none reaches 72.*

Two facts about the known configurations accompany the main result. The first is an input to its proof:

Theorem 2. *For each $X \in \{D_5, L_5, Q_5, R_5\}$, the polytope $P_X = \{w \in \mathbb{R}^5 : \langle x, w \rangle \leq 1 \ \forall x \in X\}$ is compact with $\max_{w \in P_X} |w|^2 = 5/4$, attained exactly at 32 vertices (the deep holes); consequently no 41st point can be adjoined to X .*

The second is not used in the proof but is of independent interest:

Theorem 3. *Each of D_5, L_5, Q_5, R_5 is infinitesimally jammed, hence jammed. The results for Q_5 and R_5 are new; those for D_5, L_5 reproduce [4].*

Szöllősi explicitly raised the rigidity of Q_5 as a question; the question matters because a flexible optimal configuration would open a deform-and-augment route towards $\kappa(5) \geq 41$.

We emphasize what is *not* claimed: Theorems 2 and 3 are statements about the known configurations and imply nothing about hypothetical unrelated 41-point codes. Theorem 1 says nothing about $\kappa(6)$ itself: the E_6 configuration gives 72 kissing spheres in dimension six, while the exact value of $\kappa(6)$ remains unknown, the current bounds being $72 \leq \kappa(6) \leq 77$ [8]; our theorem only bounds configurations that contain Q_5 or R_5 . The bound 70 is likely not sharp (we construct 64-point extensions and conjecture 64 is optimal).

2 Deep holes: no 41st point

Proof of Theorem 2. Write $X = \{x_1, \dots, x_{40}\}$. Then $\sum_i x_i = 0$ exactly and X spans $\mathbb{R}^5$, so the origin is interior to $\text{conv}(X)$ and P_X is compact. Every vertex of P_X is the unique solution of $\langle x_i, w \rangle = 1$ for five linearly independent constraints; enumerating all $\binom{40}{5} = 658,008$ subsets in exact integer arithmetic yields the complete vertex lists: 42/50/92/100 vertices for $D_5/L_5/Q_5/R_5$, with maximal squared norm $5/4$ attained at exactly 32 vertices in each case (for D_5 these are the deep holes of [7, §3]; for Q_5, R_5 we confirm exactly the reflection structure described there). Since $|\cdot|^2$ is strictly convex, its maximum over P_X is attained only at vertices; a 41st point x would satisfy $x \in P_X$ and $|x|^2 = 2 > 5/4$. $\square$

For later use, define the *hole graph* of X : vertices the 32 deep holes, edges the pairs with $\langle w, w' \rangle \leq 1/4$ (the maximum inner product between distinct holes is $23/20$ for Q_5, R_5 and $3/4$ for D_5, L_5). In the normalization of [7, §3] the holes lie at squared norm 2 and the threshold is $2/5$; our holes and threshold are the same up to the factor $8/5$, and for unit vectors the threshold is the $1/5$ of Lemmas 7–8.

Lemma 4. *The clique number of the hole graph is 16 for D_5 and L_5 , and 12 for Q_5 and R_5 .*

Proof. Exact computation (branch-and-bound over the 32-vertex graphs; verified by two algorithms: a library Bron–Kerbosch enumeration and an own branch-and-bound). For D_5 the 16-cliques are the half-families $S_{\text{even}}, S_{\text{odd}}$ of [7, §3]. $\square$

3 Six-dimensional extensions

Lemma 5 (Reduction). *Let $Y \subset \mathbb{R}^6$ be a kissing configuration (squared norm 2) containing a similarity copy Z of $X \in \{Q_5, R_5\}$. Then:*

1. *Z is central, at ambient scale, and $Z = Y \cap H$ for the hyperplane H it spans; we may take $H = \{y_6 = 0\}$ and $Z = X \times \{0\}$.*
2. *Every other point of Y has the form $y = (u, h)$ with $u \in P_X$, $h^2 \geq 3/4$.*
3. *Any two such points with $h > 0 > h'$ are automatically compatible.*
4. *Consequently $|Y| = 40 + |Y^+| + |Y^-|$, where $Y^\pm$ are the layers $\{h \gtrless 0\}$, and each layer is a cap set: points $y_i = (u_i, h_i)$, $u_i \in P_X$, $h_i = \sqrt{2 - |u_i|^2}$, pairwise $\langle u_i, u_j \rangle + h_i h_j \leq 1$.*

Proof. (1) X has affine rank 5, so Z spans a hyperplane $H = \{\langle y, n \rangle = c\}$, $|n| = 1$; all points of Z lie on the ambient sphere S . Since X affinely spans $\mathbb{R}^5$ its circumsphere is unique, and the similarity carries it to the unique circumsphere of Z inside H , namely $H \cap S$ with center cn and squared radius $2 - c^2$; hence $Z = cn + \lambda \cdot (\text{orthogonal image of } X)$ with $\lambda^2 \cdot 2 = 2 - c^2$. X has 240 contact pairs at inner product exactly 1; their images satisfy $\langle y, y' \rangle = \lambda^2 \cdot 1 + c^2 = 1 + c^2/2$, which respects the kissing constraint only if $c = 0$. A point $(u, 0) \in Y \setminus Z$ would satisfy $u \in P_X$, hence $|u|^2 \leq 5/4 < 2$ by Theorem 2 — impossible for a point of squared norm 2. (2) Compatibility with every $(x, 0)$ gives $u \in P_X$; $|u|^2 \leq 5/4$ gives $h^2 = 2 - |u|^2 \geq 3/4$. (3) $\langle y, y' \rangle = \langle u, u' \rangle + hh' \leq |u||u'| - \sqrt{3/4}\sqrt{3/4} \leq 5/4 - 3/4 = 1/2 \leq 1$. (4) Immediate from (1)–(3); conversely any cap set and its mirror assemble with $X \times \{0\}$ into a valid configuration. $\square$

Theorem 6. *There exist 64-point six-dimensional kissing configurations with Q_5 (resp. R_5) as central cross section: the cross-section together with a 12-point hole clique placed at heights $\pm\sqrt{3}/2$. (Exact certificates accompany the paper.)*

We now bound cap sets. Write $M(X)$ for the maximum size of a cap set; by Lemma 5, Theorem 1 is equivalent to $M(Q_5), M(R_5) \leq 15$.

Lemma 7 (Norm coupling). *Let $y = (u, h)$, $y' = (u', h')$ be distinct compatible cap points with $u, u' \neq 0$. Then $\langle \hat{u}, \hat{u}' \rangle \leq 1/5$ for the unit vectors $\hat{u} = u/|u|$, $\hat{u}' = u'/|u'|$, with equality only if $|u|^2 = |u'|^2 = 5/4$.*

Proof. Compatibility gives $\langle \hat{u}, \hat{u}' \rangle \leq (1 - hh')/(|u||u'|)$. With $a = h$, $b = h'$, $|u| = \sqrt{2 - a^2}$, $|u'| = \sqrt{2 - b^2}$, and $a, b \in [\sqrt{3}/2, \sqrt{2})$, the claim reduces to $G(a, b) := \sqrt{(2 - a^2)(2 - b^2)} - 5(1 - ab) \geq 0$. If $ab \geq 1$ this is trivial. If $ab < 1$, both sides of $\sqrt{(2 - a^2)(2 - b^2)} \geq 5(1 - ab)$ are nonnegative, and squaring gives the equivalent

$$2(a^2 + b^2) + 24(ab)^2 - 50ab + 21 \leq 0.$$

Set $x = ab \in [3/4, 1)$ and write $b = x/a$; the constraints $a, b \geq \sqrt{3}/2$ confine a to $[\sqrt{3}/2, 2x/\sqrt{3}]$. The function $\varphi(a) = a^2 + x^2/a^2$ is strictly convex for $a > 0$ ($\varphi''(a) = 2 + 6x^2/a^4 > 0$), so its maximum over this interval is attained at an endpoint, and by the symmetry $a \leftrightarrow x/a$ both endpoints give the same value $\varphi(\sqrt{3}/2) = 3/4 + 4x^2/3$. Hence the left side is at most $(80/3)x^2 - 50x + 45/2$, a quadratic with roots exactly $x = 3/4$ and $x = 9/8$, so it is ≤ 0 on $[3/4, 9/8] \supset [3/4, 1)$. Equality forces $x = 3/4$ and $a = b = \sqrt{3}/2$, i.e. $|u|^2 = |u'|^2 = 5/4$; tracing back, equality in the lemma also requires $\langle u, u' \rangle = 1 - hh'$. In angular variables $|u| = \sqrt{2} \sin \theta$, $h = \sqrt{2} \cos \theta$ the claim reads $10 \cos \theta \cos \theta' + 2 \sin \theta \sin \theta' = 6 \cos(\theta - \theta') + 4 \cos(\theta + \theta') \geq 5$ on $(0, \theta^*]^2$, $\sin^2 \theta^* = 5/8$: for fixed $\theta + \theta'$ the left side decreases in $|\theta - \theta'|$, so the minimum is on the edge $\theta' = \theta^*$, where $g(\theta) = 6 \cos(\theta - \theta^*) + 4 \cos(\theta + \theta^*)$ is concave with $g(\theta^*) = 5 < g(0)$. $\square$

Two degenerate remarks: the pole point $(0, \sqrt{2})$ is incompatible with every other cap point ($g = \sqrt{2}h' \geq \sqrt{3/2} > 1$), so a cap set of size ≥ 2 has all $u_i \neq 0$; and two cap points never share a direction ($\langle \hat{u}, \hat{u}' \rangle = 1 > 1/5$). Thus in a cap set of size $N \geq 2$ the $\hat{u}_i$ are N distinct unit vectors in S^4 with pairwise inner products $\leq 1/5$.

Lemma 8 (LP bound, exact certificate). *Any N unit vectors in $\mathbb{R}^5$ with pairwise inner products $\leq 1/5$ satisfy $N \leq 16$.*

Proof. This is a case of Levenshtein's bound [11]; we include the explicit certificate since its equality analysis is used below. Let G_k denote the Gegenbauer polynomials for S^4 (parameter $\lambda = 3/2$) normalized by $G_k(1) = 1$, positive definite on S^4 (Schoenberg [9]): $G_1(t) = t$, $G_2(t) = (5t^2 - 1)/4$, $G_3(t) = (7t^3 - 3t)/4$. One has the exact identity

$$F(t) := 1 + (30/7)G_1(t) + (25/4)G_2(t) + (125/28)G_3(t) = (125/16)(t + 3/5)^2(t - 1/5),$$

with $f_0 = 1$, $f_1 = 30/7$, $f_2 = 25/4$, $f_3 = 125/28$, all coefficients positive, $F \leq 0$ on $[-1, 1/5]$, and $F(1) = 16$. For unit vectors $v_1, \dots, v_N$ with pairwise $t_{ij} \leq 1/5$, using $\sum_{i,j} G_k(t_{ij}) \geq 0$ (Schoenberg [9]),

$$N^2 = \sum_{i,j} G_0(t_{ij}) \leq \sum_k f_k \sum_{i,j} G_k(t_{ij}) = N F(1) + \sum_{i \neq j} F(t_{ij}) \leq 16N. \quad \square$$

Remark 9 (Sharpness). The bound is attained by the 5-demihypercube: the 16 vectors $(\pm 1, \dots, \pm 1)/\sqrt{5}$ with an even number of minus signs, whose pairwise inner products lie in $\{1/5, -3/5\}$. This code attains Levenshtein's bound [11] and is the unique $(5, 16, 1/5)$ spherical code up to isometry [12, Appendix A]. This is exactly half of the deep-hole family of D_5 — whence $M(D_5) = 16$ and the classical E_6 extension. The linear program alone therefore cannot give 15: the hole structure of Q_5, R_5 in the next lemma is essential.

Lemma 10 (Equality-case rigidity). *$M(Q_5) \leq 15$ and $M(R_5) \leq 15$.*

Proof. Suppose a cap set over $X \in \{Q_5, R_5\}$ had 16 points. Then equality holds throughout the proof of Lemma 8 for the $\hat{u}_i$ (the equality analysis of Delsarte–Goethals–Seidel [10]). In particular $\sum_{i \neq j} F(t_{ij}) = 0$ with every term ≤ 0 , so every t_{ij} lies in the root set $\{-3/5, 1/5\}$; and $\sum_{i,j} G_k(t_{ij}) = 0$ for $k = 1, 2, 3$ (all $f_k > 0$). For $k = 1$ this reads $0 = \sum_{i,j} G_1(t_{ij}) = \sum_{i,j} \langle \hat{u}_i, \hat{u}_j \rangle = |\sum_i \hat{u}_i|^2$, so $\sum_i \hat{u}_i = 0$, and pairing with $\hat{u}_i$: $1 + \sum_{j \neq i} t_{ij} = 0$. If n_i pairs equal $1/5$ (and $15 - n_i$ equal $-3/5$), then $1 + n_i/5 - 3(15 - n_i)/5 = 0$, i.e. $n_i = 10$ for every i : each point has ten partners at inner product exactly $1/5$.

By the strictness in Lemma 7, each such pair forces both endpoints to have $|u|^2 = 5/4$ exactly; since $n_i > 0$ for all i , all sixteen u_i have maximal norm, hence (Theorem 2, sphere-touching) all are deep-hole vertices, pairwise distinct. Their compatibility at $h_i = h_j = \sqrt{3}/2$ reads $\langle u_i, u_j \rangle \leq 1/4$: a 16-clique in the hole graph, contradicting Lemma 4. $\square$

Proof of Theorem 1. By Lemma 5, $|Y| = 40 + |Y^+| + |Y^-| \leq 40 + 15 + 15 = 70 < 72$. $\square$

Remark 11 (Calibration). Applied to D_5 or L_5 , the same chain yields no contradiction — their hole graphs do contain 16-cliques — and Lemmas 5–8 give $|Y| \leq 72$, exactly attained by the E_6 and Leech six-dimensional configurations. The argument is tight where it must be.

Conjecture 12. $M(Q_5) = M(R_5) = 12$; equivalently, the 64-point configurations of Theorem 6 are the largest six-dimensional kissing configurations with Q_5 or R_5 cross sections. (Supported by the exact hole-clique number 12 and by a multistart penalty minimization, 200 restarts per case, script `experiments/conjecture31_stageA.py`, which reaches 12 points to machine precision and finds no 13-point cap set; this probe is not part of the verified claims. Closing 12 versus 15 requires more than the LP bound, by Remark 9.)

4 Jamming

We record the rigidity of the four configurations. This is not used above; it answers Szöllősi’s question and closes the deform-and-augment route mentioned in the introduction. Let $X = \{x_1, \dots, x_{40}\}$ be one of the four configurations. An *infinitesimal flex* is $(v_1, \dots, v_{40})$, $v_i \in \mathbb{R}^5$, with $\langle x_i, v_i \rangle = 0$ for all i and $g_{ij}(v) := \langle x_i, v_j \rangle + \langle x_j, v_i \rangle \leq 0$ for every contact $\{i, j\}$. Trivial flexes are $v_i = Ax_i$ with A antisymmetric; they form a 10-dimensional space since X spans $\mathbb{R}^5$. X is *infinitesimally jammed* if every flex is trivial; infinitesimally jammed implies jammed [4, Thm. 2.2] (Connelly; Roth–Whiteley).

Proof of Theorem 3. Two exact certificates per configuration.

(i) *A strictly positive self-stress:* weights $\omega_e > 0$ on the 240 contacts with $s_i := \sum_{j: \{i,j\} \in E} \omega_{ij} x_j$ parallel to x_i for every i . For D_5 and L_5 the uniform stress $\omega \equiv 1$ works, with $s_i = 6x_i$. For Q_5 the uniform stress provably fails; a valid stress takes the two values $20/21$ (on 210 contacts) and $4/3$ (on 30), with $s_i = \lambda_i x_i$, $\lambda_i \in \{40/7, 44/7\}$. For R_5 the values are $200/207$ ($\times 210$), $280/207$ ($\times 18$), $220/207$ ($\times 12$), with $\lambda_i \in \{400/69, 440/69, 140/23\}$. Verification is a finite exact computation. For any flex v ,

$$\sum_e \omega_e g_e(v) = \sum_i \langle s_i, v_i \rangle = \sum_i \lambda_i \langle x_i, v_i \rangle = 0,$$

and since every $\omega_e > 0$ and every $g_e(v) \leq 0$, all $g_e(v) = 0$: every flex preserves every contact to first order.

(ii) *The equality kernel is trivial:* the stacked linear system of the 40 tangency equations and 240 contact equalities in the 200 unknowns has rank exactly 190 over $\mathbb{Q}$ (computed by fraction-free Bareiss elimination, and independently via a mod- p argument: $\text{rank}_{\mathbb{F}_p} \leq \text{rank}_{\mathbb{Q}}$ together with the 10-dimensional space of rotation flexes in the kernel). Hence the kernel is exactly the trivial flexes, and X is infinitesimally jammed. $\square$

5 Methodology, verification, and data

All verdict-relevant computations are exact: configurations are stored with exact coordinates (rational; the 64-point extensions carry $\sqrt{3}/2$ symbolically); an exact-arithmetic checker (canonical multiquadratic representation with a decidable sign test, no floating point on any accepting path) verifies all norm and inner-product claims; the vertex enumeration of Section 2 runs in integer arithmetic with proven overflow bounds; stresses carry rational certificates; ranks are recomputed exactly by two independent implementations and clique numbers by two algorithms. Three of the computational claims are checked by two independent computations: the jamming certificates of Theorem 3 (positivity and parallelism of the stress and the rank 190; one implementation uses dense integer rows and a λ -multiple test, the other sparse rows assembled from the contact graph, a 2×2 -minor test, different primes and an exact Bareiss rank), the vertex enumeration of Theorem 2 (a float-prefiltered enumeration with exact re-verification, and a tolerance-free integer enumeration), and the clique numbers of Lemma 4 (a library Bron–Kerbosch enumeration and an own branch-and-bound). The Gegenbauer identity of Lemma 8 is checked by a single symbolic expansion; the 64-point configurations of Theorem 6 by the exact checker on the shipped coordinates and, separately, by rebuilding them from a 12-clique of the hole graph. Lemmas 5, 7 and 10 are proved in the text and use only the verified inputs above. The repository <https://github.com/smitterer/kissingnumbers> (cite as github.com/smitterer/kissingnumbers, release v1.1) contains all configurations, certificates, both implementations, and an experiment log; a single command re-runs every check against a table of expected values.

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References

- [1] A. Korkine, G. Zolotareff, Sur les formes quadratiques, *Math. Ann.* **6** (1873), 366–389.
- [2] J. Leech, Five dimensional non-lattice sphere packings, *Canad. Math. Bull.* **10** (1967), 387–393.
- [3] J.H. Conway, N.J.A. Sloane, What are all the best sphere packings in low dimensions?, *Discrete Comput. Geom.* **13** (1995), 383–403.
- [4] H. Cohn, Y. Jiao, A. Kumar, S. Torquato, Rigidity of spherical codes, *Geom. Topol.* **15** (2011), 2235–2273.

- [5] H.D. Mittelmann, F. Vallentin, High-accuracy semidefinite programming bounds for kissing numbers, *Exp. Math.* **19** (2010), 175–179.
- [6] F. Szöllősi, A note on five dimensional kissing arrangements, *Math. Res. Lett.* **30** (2023), no. 5, 1609–1615; arXiv:2301.08272.
- [7] H. Cohn, I. Rajagopal, Variations on five-dimensional sphere packings, *Discrete Comput. Geom.* (2026), doi:10.1007/s00454-026-00841-x; arXiv:2412.00937.
- [8] D. de Laat, N. Leijenhorst, W.H.H. de Muinck Keizer, Optimality and uniqueness of the D_4 root system, arXiv:2404.18794.
- [9] I.J. Schoenberg, Positive definite functions on spheres, *Duke Math. J.* **9** (1942), 96–108.
- [10] P. Delsarte, J.M. Goethals, J.J. Seidel, Spherical codes and designs, *Geom. Dedicata* **6** (1977), 363–388.
- [11] V.I. Levenshtein, On bounds for packings in n -dimensional Euclidean space, *Dokl. Akad. Nauk SSSR* **245** (1979), no. 6, 1299–1303 (Russian); English translation: *Soviet Math. Dokl.* **20** (1979), no. 2, 417–421.
- [12] H. Cohn, A. Kumar, Universally optimal distribution of points on spheres, *J. Amer. Math. Soc.* **20** (2007), no. 1, 99–148.